

Hassan Abbas

Economist | Econometrics & Data Science | Economic Impact & Scenario Analysis
76133 Karlsruhe, Germany | +49 172 3793893 | hassanabbas.baig@gmail.com | linkedin.com/in/has-san-ab-bas

Professional Profile

Applied economist with central-banking and research experience across macroeconomic forecasting, econometric modelling, scenario analysis and policy evaluation. Experienced in translating economic questions into empirical strategies using time-series, panel and cross-sectional econometrics, semi-structural models, accounting frameworks, machine learning and large economic and financial datasets. Combines economic modelling with hands-on data engineering, including construction of new indicators and analytical datasets from granular, unstructured and geospatial sources.

Selected Highlights

- Prepared and integrated real, fiscal, external, monetary and financial-sector forecasts into consistent baseline and alternative medium-term macroeconomic scenarios.
- Extended forecasting with QPM/IRIS, SVAR and machine-learning methods, including high-frequency GDP nowcasting and models for inflation, trade in services and workers' remittance inflows.
- Integrated firm financial statements, daily tax collections, business-level FDI and weekly credit flows into real, fiscal and external-sector forecasts.
- Estimated heterogeneous bank-lending responses to monetary-policy shocks using 42 months of eCIB data covering 72 institutions, ~3 million borrowers and ~5 million loans per month.
- Helped conceive and establish the SBP-IBA Survey Research Centre, translating the central bank's need for systematic expectations data into a permanent nationwide research platform.
- Built a reusable multilingual knowledge-discovery platform over 2.8+ million news articles spanning 25 years, integrating a persistent Knowledge Graph with local LLM/GraphRAG-based retrieval.

Professional Experience

Economist, Monetary Policy Department, State Bank of Pakistan

Mar 2025–Mar 2026

Karachi, Pakistan

- Prepared the Medium-Term Macroeconomic Framework by integrating real, fiscal, external, monetary and financial-sector forecasts into consistent baseline and alternative scenarios for medium-term policy assessment.
- Assessed inflation, output, financial conditions and transmission of policy and external shocks using QPM/IRIS, SVARs, econometric forecasting and scenario analysis in MATLAB with macroeconomic and supervisory datasets.
- Evaluated structural reforms, sovereign-financing developments, financial regulation and geopolitical shocks by combining model-based scenarios with sectoral, financial and macroeconomic evidence.
- Reviewed a World Bank-supported climate-finance facility by assessing a flood-risk methodology linking hazard probabilities and microfinance-institution exposures to estimated losses, liquidity requirements and facility activation.
- Coordinated Medium-Term Unit outputs across sectoral forecasts and technical review while remaining directly involved in modelling, scenario design and quantitative analysis.

Student Assistant (HiWi), Chair of International Economic Policy, University of Freiburg

Sep 2023–Mar 2025

Freiburg, Germany

- Developed reproducible R and Stata workflows for microeconometric research using survey and other structured datasets; reviewed literature, data and empirical specifications across research projects.

Senior Analyst, Monetary Policy Department, State Bank of Pakistan

Sep 2018–Sep 2021

Karachi, Pakistan

- Developed real, fiscal and external-sector forecasts for the Medium-Term Macroeconomic Framework and integrated sectoral projections into baseline and alternative policy scenarios.
- Linked granular and high-frequency indicators to macroeconomic forecasts, including firm financial statements, tax collections, business-level FDI and weekly credit data.
- Built high-frequency GDP nowcasting workflows using LASSO and applied SVM and clustering methods in Python and MATLAB to inflation, trade in services and remittance forecasting.
- Quantified effects of tax-efficiency reforms on tax revenues, manufacturing output and firm size using 12 years of financial statements covering publicly listed Pakistani firms.
- Developed R/Shiny and Power BI dashboards for bilateral trade analysis at up to HS-6 product level; produced forecast-consistency checks, alternative scenarios and quantitative briefing material for policy deliberations.

Senior Analyst, International Relations Division, Research Department, State Bank of Pakistan

Jul 2016–Sep 2018

Karachi, Pakistan

- Prepared analytical briefs and coordinated economic, financial and policy inputs for international engagements with the IMF, World Bank and regional central banks.

Analyst / Senior Analyst, Research Department, State Bank of Pakistan

Jan 2009–Jun 2016

Karachi, Pakistan

- Conducted applied research on monetary-policy shocks, financial stability, inflation expectations, international trade, agricultural

finance and taxation using administrative, supervisory, firm, survey and macroeconomic data.

- Estimated heterogeneous bank-lending responses to monetary-policy shocks across easing and tightening cycles using panel/cohort fixed-effects and time-effects models on eCIB data, complemented by QGIS spatial analysis.
- Served as SBP focal point and technical lead in establishing the SBP–IBA Survey Research Centre, taking the Consumer Confidence Survey from research and sampling design through piloting and implementation into a permanent nationwide platform.
- Designed survey methodology and sampling; cleaned and geographically mapped 3.5 million records; translated methodology into operational and system requirements spanning CRM, telephony, data capture, monitoring and reporting; worked with developers and vendors through implementation.
- Contributed quantitative analysis to the State Bank–IMF macroprudential policy technical team using supervisory bank balance-sheet data, econometric specifications and scenario assumptions.

Quantitative & Data Science Toolkit

Econometrics & Modelling	QPM/IRIS; SVAR; time-series forecasting; panel and cross-sectional econometrics; event studies; fixed-effects methods; survey design; scenario and sensitivity analysis.
Machine Learning & AI	LASSO; SVM; clustering; NLP; embeddings/vector search; Knowledge Graphs; GraphRAG and local LLM workflows.
Programming	Python (pandas, NumPy, SciPy, scikit-learn); R; Stata; MATLAB; EViews; SQL; Excel/VBA.
Data Engineering	Git; Docker; Conda; REST APIs; relational, graph and vector databases; Neo4j; Qdrant; reproducible pipelines.
Visualisation	Power BI; Tableau; R/Shiny; QGIS; Quarto; HTML; GEE/NASA-MODIS.
Data Experience	eCIB; supervisory banking; firm financial statements; household/business surveys; trade/BoP; IMF-IFS, Haver, World Bank, Eurostat, OECD; Bloomberg; news/text; geospatial, climate and satellite data.

Master's Thesis Research

University of Freiburg

Mar 2026–Present

- Investigating long-term institutional change following natural disasters by combining 25 years of multilingual news with 26 years of household/individual survey data, structured events and geospatial evidence.
- Built a reusable multilingual knowledge-discovery platform over 2.8+ million articles using Neo4j, Qdrant, embeddings, multilingual NLP and local LLM/GraphRAG workflows.
- Harmonised 1.4+ million individual- and household-level observations and linked news, survey, event, climate and satellite sources to construct spatially and temporally aligned measures for subsequent microeconomic analysis.

Education

MSc Economics, University of Freiburg *Freiburg, Germany*

Expected 2026

MA Economics, University of Agriculture Faisalabad *Faisalabad, Pakistan*

2003–2005

Professional Training

IMF: Macroeconomic Forecasting | Advanced Econometric Forecasting | Financial Programming and Policies
Deutsche Bundesbank, Frankfurt: Expert Panel — Big Data and Central Banking